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P-XX

Two-Layer Topological Mapping for Energy-Aware Arm and Base Selection on a Dual-Gantry Quad-Arm Ceiling Robot

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One topological representation, two layers:
the room the robot sees, and the reach each arm owns, priced in energy.

Social Background

An aging population is driving demand for assistive robots that work **inside the home**: fetching, tidying, delivering.

Homes are small, cluttered, and **shared with residents**, so a floor-driving mobile manipulator gets in the way.

Putting the robot on the ceiling keeps the floor free: it assists from above without blocking the occupant.

Platform. Two ceiling rails (gantries), **two Kinova Gen3 Lite arms each** → four arms. Each rail adds a prismatic axis d and a revolute axis θ that its two arms *share*, so every arm is an **8-DOF** chain $\mathbf{q} = [d, \theta, q_1 \dots q_6]$ and moving the base moves two arms at once. Sensing is **RGB-D only**: no LIDAR.

Problems

- **P1 — Two coupled decisions.** Each pick asks *which arm* acts *and where its base stops*. Moving the heavy base is a dominant energy cost.
- **P2 — Maps assume a fixed base.** Classic reachability/capability maps are built for one arm on a stationary base, so they never see the workspace that base motion creates — and a cell stores only *whether* it is reachable, not at what cost or from which configuration.
- **P3 — Occupancy is not a map of the room.** A raw octomap carries no topology, and in a single live layer the fixed structure of the room and a person walking through it are indistinguishable, so the scene is never reproducible between runs.

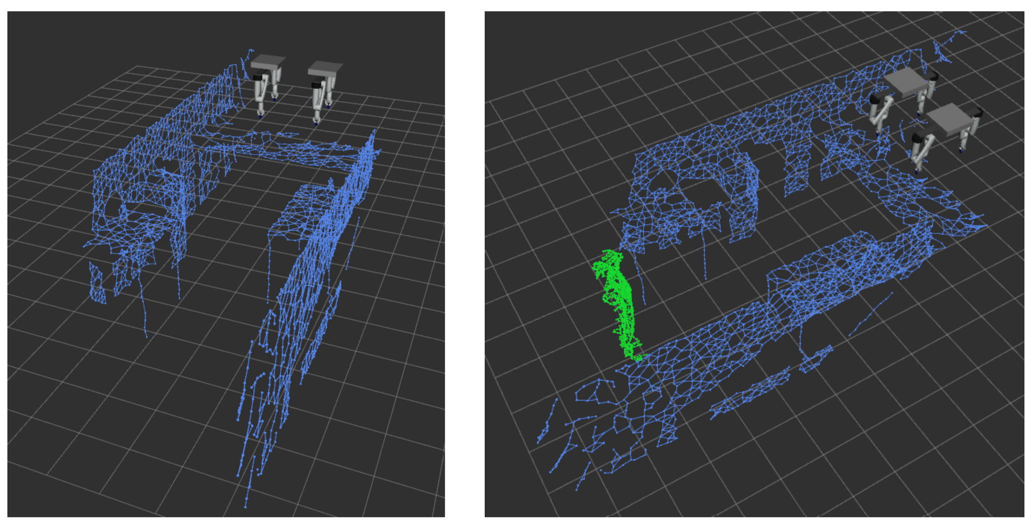
Core Idea: One GNG, Two Layers

The *same* **Growing Neural Gas** representation is fitted twice: to **what the cameras see** (environment layer) and to **what each arm can reach** (capability layer). Planning collides against the first; energy-aware selection scores candidates on the second. Nodes are the common currency: a node is a place, plus whatever the robot needs to know about it.

Layer 1: Environment GNG from RGB-D

- Two overhead RGB-D cameras are the *only* exteroceptive sensor (**no LIDAR**); their extrinsics are known from the simulated scene and published as `world`→camera static transforms. On hardware the same transforms come from a **ChArUco** board; the calibration path is prepared, not yet exercised.
- `depth_cloud` deprojects each depth image into world-frame points **without any segmentation gate**, so the map keeps building even when the detector is blind.
- A GNG is fitted to that cloud: nodes tile the observed geometry and competitive Hebbian edges give the scene a *topology*, not just occupancy.
- **Static sub-layer:** captured once on a cleared scene and frozen to a file: a backbone that is *identical every run*. **Dynamic sub-layer:** live points within `bg_dist` of a static node are subtracted, so the live map carries only the genuinely moving remainder (a person, a displaced object).
- Both are unioned into the planning scene, so MoveIt 2 plans against fixed structure and dynamic obstacles at once. Object identity is kept separate and label-free at the geometry level: tracks are associated by **Hungarian matching** with an **N-frame confirmation** and voting hysteresis, and only the confirmed target is carved out of the collision map.

*The two overhead RGB-D views that are the entire input: camera 1 (left) and camera 2 (right), which also sees the person: fixed structure for the static sub-layer, the person and moved objects for the dynamic one. Everything runs in **NVIDIA Isaac Sim**, a digital twin of our trailer living laboratory built from the same URDF as the real robot.*



*The resulting two-layer map (RViz, same viewpoint). **Left:** the frozen static backbone: walls, tables, the fixed structure of the room. **Right:** the same backbone with the dynamic remainder after subtraction: here a person who entered the cell, the only thing the live layer has to carry.*

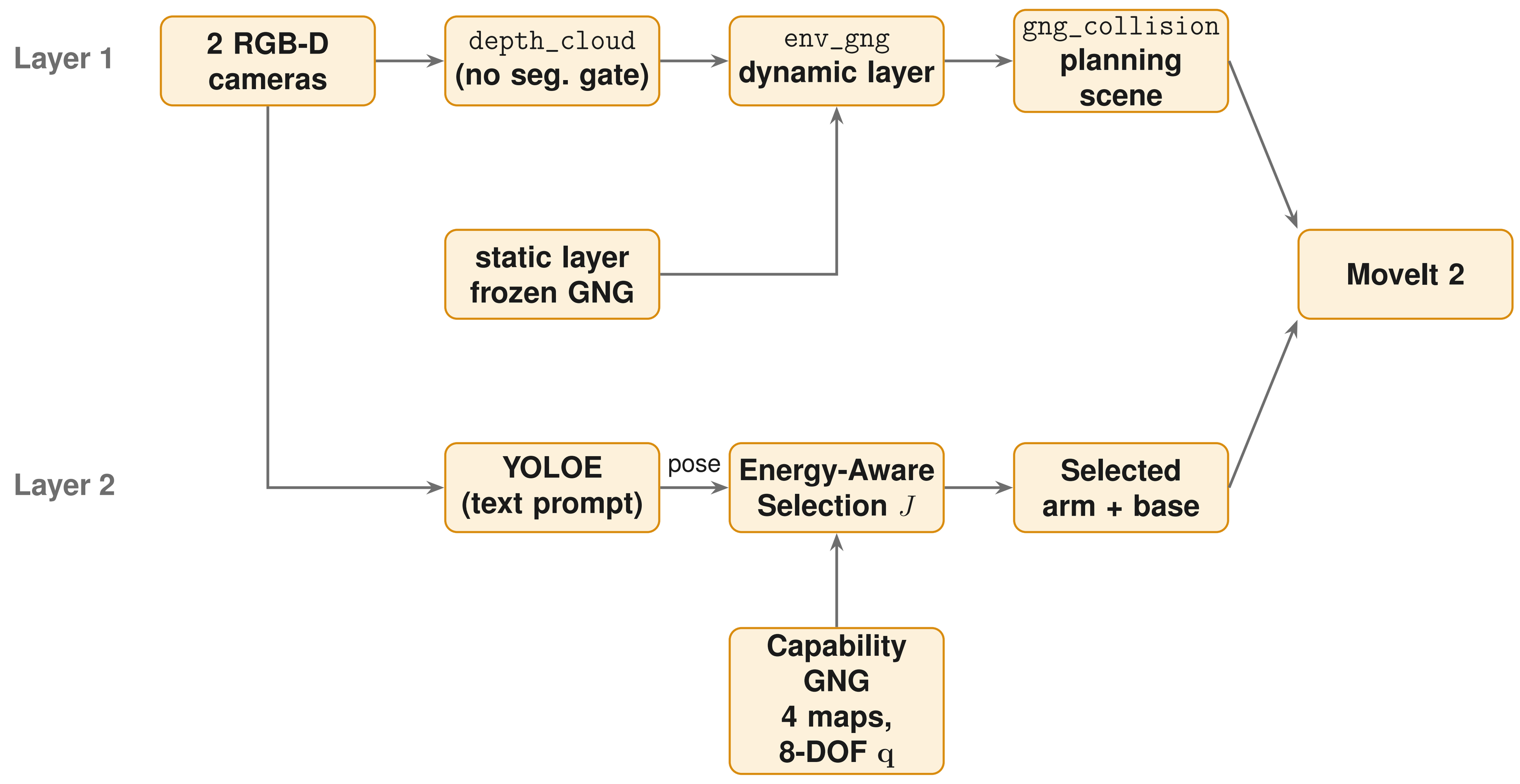
Layer 2: Base-Aware Capability GNG

- Sample the **full 8-DOF** $\mathbf{q} = [d, \theta, q_1 \dots q_6]$ inside joint limits, take forward kinematics (Pinocchio), record manipulability $w = \sqrt{\det(JJ^T)}$.
- Each node stores $[\,\mathbf{x} \mid \mathbf{q}\,]$: a reachable point *and* a representative configuration reaching it, **rail placement included**. Winner search uses the *task* part only, so a node's $\mathbf{q}$ stays one valid posture, not an average of incompatible base positions.
- **Boundary seeding**: a shell of nodes is pinned on the measured reach surface *before* the interior grows, keeping the map faithful at the workspace edge. One map per arm → four maps.

Capability layer of one arm (RViz): nodes coloured by manipulability plus the pinned boundary shell. The elongated extent is the rail sweeping x.

Map size, per arm: 80,000 FK samples → **2915 nodes** (600 pinned on the boundary shell), 14,547 edges, median node spacing **0.090 m**, and a full **8-DOF** $\mathbf{q}$ stored on every node.

System and Pipeline



Geometry (Layer 1) is decoupled from identity and decision (Layer 2): the environment map builds from `depth_cloud` alone, so it survives a detector that sees nothing, while selection queries the capability maps.

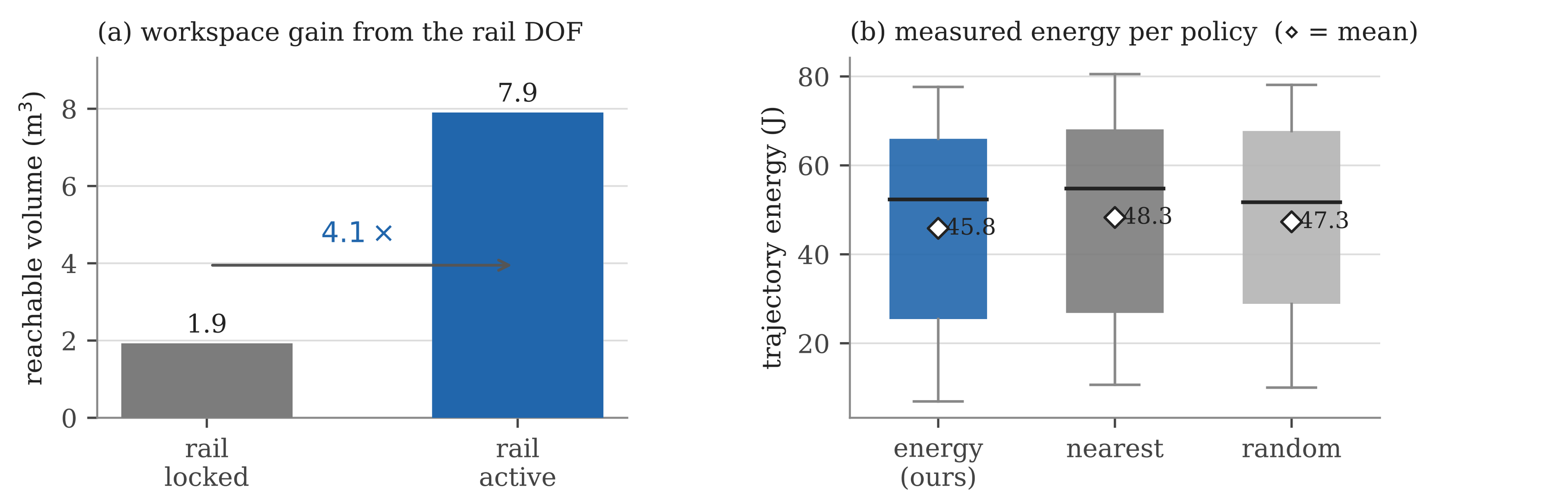
Energy-Aware Selection Cost J

$$J = w_{gl}\frac{\Delta_{lin}}{\rho_{gl}} + w_{gr}\frac{\Delta_{rot}}{\rho_{gr}} + w_{arm}\frac{\Delta_{arm}}{\rho_{arm}} + w_{dist}\frac{e}{\rho_{dist}} - w_{manip}\frac{m}{\rho_{manip}}$$

- $\Delta_{lin}, \Delta_{rot}, \Delta_{arm}$: rail-linear, rail-rotation and summed arm-joint travel from the current state; e: tool-to-object distance; m: manipulability, entering *negatively*. Each term is divided by a reference ρ (its median), so a weight reads as a priority, not a unit artefact.
- Weights are **fit by OLS** against the *measured* mechanical energy of 168 executed picks, not hand-tuned ($R^2 = 0.857$, Spearman$(J, E) = 0.909$): Δ_{lin} **27.16**, e 12.78, Δ_{rot} 3.08, m 2.95, Δ_{arm} 0.09. Rail travel dominates once rail friction is modelled; the calibration *recovers* the premise that moving the base is the dominant cost.
- **Selection.** Pool capability-layer nodes within a spacing-scaled radius of the target *from all four arms*, score with J, and try them **round-robin across arms**. Each candidate seeds IK from its own $\mathbf{q}$; MoveIt 2 plans against the environment layer; the first collision-free plan wins. The search is **plan-only**: the arm never moves while deciding.

Experiments and Results

Workspace gain: the rail is worth $4\times$ the reachable volume. Rail locked (6 arm DOF) vs. rail sampled over its full travel, at matched sample *density*: reachable volume grows **1.920→7.899 m^3** at 0.05 m voxels. **Boundary seeding** cuts the node-hull shortfall against the true FK reach surface from **0.221 m to 0.008 m**.



(a) Reachable volume, rail locked vs. active. (b) Measured trajectory energy per multi-arm policy (◇ = mean).

Selection policies, 60 target poses, plan-only, mean/median over successful picks, under an assumed rail friction model.

Mode	Success	Rail lin. (m)	Energy (J)
energy (ours)	**56/60 (93.3%)**	**0.97**/1.01	**45.84**/52.34
nearest	56/60 (93.3%)	1.19/1.41	48.28/54.79
random	56/60 (93.3%)	1.17/1.28	47.31/51.72
fixed (best, arm_4)	39/60 (65.0%)	1.14/1.21	49.79/55.78

Fixed-arm energies are computed over that arm's own smaller reachable set, so they are not comparable at equal coverage.

- **Coverage:** every multi-arm policy reaches **93.3%** plan success vs. **63–65%** for any single fixed arm.
- **Energy:** ours has the **lowest mean measured energy** of the multi-arm policies (**45.84 J** vs. 48.28 J nearest, -5.1%), winning **57–61%** of same-target pairwise comparisons, a real but *partial* win: medians are closer and the distributions overlap.
- **The intended trade appears:** ours spends *more* arm travel to avoid costlier rail travel; one logged pick took arm_3 (0.64 m rail, 42.4 J) over the nearest arm_4 (1.15 m rail, 72.5 J): **30.1 J** saved.

End-to-End Validation

Full pipeline: perception → environment layer → energy-aware selection → IK → planning → real Isaac physics execution, on 7 objects × 5 trials: **35/35 (100%)** picks, mean time-to-pick **39.6 s**.

Perception: YOLOE vs. ground truth (8 objects, 483 frames): **100%** detection, **0.023 m** mean localization error. **Reachability screening:** the capability layer's nearest-node test classifies reachable vs. unreachable objects **100% (8/8)**.

Environment layer (measured live in Isaac): static backbone **1800 nodes, 3569 edges**, frozen once on the empty scene; the live layer fills to its **800-node** cap (the objects on the tables plus the person are all new relative to that empty backbone) within under a minute, publishing at **~1 Hz**.

Summary and Contributions

- **One GNG, two layers**: a single topological representation for the *environment* (RGB-D only, static backbone + dynamic remainder, reproducible between runs) and for each arm's *capability*.
- A **base-aware capability layer** treating the ceiling rail as a genuine DOF: every node carries its own rail placement, posture and manipulability, accurate to **centimetres** at the workspace edge.
- An **energy-aware cost** J, calibrated against measured energy, that picks the arm *and* its full 8-DOF configuration: $\approx 4\times$ workspace gain, 93.3% vs. 63–65% plan success, lowest mean energy among multi-arm policies, 35/35 end-to-end picks.

Future work. All results are in simulation. Next: validation on the physical ceiling-rail platform, and a dynamic-scene study where the environment layer changes and the cheapest arm changes with it.

References. [1] Fritzke, NIPS 1995 (GNG). [2] Zacharias *et al.*, IROS 2007. [3] Makhal & Goins, IRC 2018 (Reuleaux). [4] Yoshikawa, IJRR 1985. [5] Sandakalum *et al.*, Humanoids 2022. [6] Shao *et al.*, arXiv:2403.19940 (MoMa-Pos). [7] Naik *et al.*, IROS 2024 (BaSeNet). [8] Wang *et al.*, ICCV 2025 (YOLOE). [9] Carpentier *et al.*, SII 2019 (Pinocchio). [10] Coleman *et al.*, JOSER 2014 (MoveIt).